

CO2-AT1-SCENARIO BASED PROGRAMS

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COURSE CODE: CSA0904

COURSE NAME: PROGRAMMING FOR JAVA

1. Suppose that a university system stores student data, where each student must be unique and can enroll in multiple courses.

(a) Enumerate three Collection interfaces suitable for this system.

(b) Design a Collection structure (diagram/representation) to store students and their enrolled courses.

(c) Write a Java program using HashSet and HashMap with Generics to:

. Add students

. Map students to courses

. Display data using Iterator

The screenshot displays the SIMATS Online Programming Lab interface. The browser address bar shows the URL: `simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=747`. The page header includes the SIMATS Engineering logo, the title "SIMATS Online Programming Lab", and the user's name "Dharshan A N" with their registration number "192525370RA".

On the left sidebar, there are two questions listed: "Q4 Employee Data Processing" (10 marks) and "Q5 Online Voting System" (10 marks). Below them, it says "5/5 answered".

The main area is titled "Your Code" and contains a Java program that implements the requirements:

```
1 import java.util.*;
2 class Main {
3     public static void main(String[] args) {
4         HashSet<String> s = new HashSet<>();
5         HashMap<String, ArrayList<String>> c = new HashMap<>();
6         s.add("John");
7         s.add("Ram");
8         s.add("Priya");
9         c.put("John", new ArrayList<>(Arrays.asList("Java", "Python")));
10        c.put("Ram", new ArrayList<>(Arrays.asList("C", "DEMS")));
11        c.put("Priya", new ArrayList<>(Arrays.asList("Java", "AI")));
12        Iterator<String> it = s.iterator();
13        while (it.hasNext()) {
14            String si = it.next();
15            System.out.println("Student: " + si);
16            System.out.println("Courses: " + c.get(si));
17        }
18    }
19 }
```

Below the code editor are "Run" and "Save" buttons.

On the right side, there is an "Input" section with a text area labeled "stdin input". Below it is an "Output" section showing the program's output:

```
Student: Priya
Courses: [Java, AI]
Student: John
Courses: [Java, Python]
Student: Ram
Courses: [C, DEMS]
```

The bottom of the image shows a Windows taskbar with the date "06-08-2026" and time "13:42".

2. Suppose that an online shopping system maintains a cart where products are added, removed, and displayed in order.

(a) List three Collection classes applicable for cart management.

(b) Represent how cart items are stored using a List-based structure.

(c) Write a Java program using ArrayList and Iterator to:

. Add items

. Remove an item

. Display all items

The screenshot displays the SIMATS Online Programming Lab interface. The browser address bar shows the URL: `simatslab.saveetha.com/practice_app/classactivity/attempt/assignment_id=747`. The page header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", and the user's name "Dharshan A N" with a profile icon. The main content area is titled "Display all items" and contains a code editor, an input field, and an output display.

Your Code

```
1 import java.util.*;
2 class Main {
3     public static void main(String[] args) {
4         ArrayList<String> c = new ArrayList<>();
5         c.add("Laptop");
6         c.add("Mouse");
7         c.add("Keyboard");
8         String r = "Mouse";
9         Iterator<String> it = c.iterator();
10        System.out.println("Cart Items:");
11        while (it.hasNext()) {
12            String item = it.next();
13            if (!item.equals(r))
14                System.out.println(item);
15        }
16    }
17 }
```

Input

stdin input

Output

Cart Items:
Laptop
Keyboard

The interface also shows a sidebar with a list of questions: "Library Management System" (10 marks), "Q4 Employee Data Processing" (10 marks), and "Q5 Online Voting System" (10 marks). The status "5/5 answered" is displayed below the questions. At the bottom of the page, there is a Windows taskbar showing the system clock as 13:42 on 06-08-2026, and the battery level at 72%.

3. Suppose that a library stores books with unique ISBN numbers and needs fast retrieval.

(a) Name three Map implementations in Java.

(b) Design a schema-like representation using Map for storing ISBN and book details.

(c) Write a Java program using HashMap to:

. Insert book details

. Retrieve a book

. Display all books

The screenshot displays the SIMATS Online Programming Lab interface. The top navigation bar includes the SIMATS ENGINEERING logo, the lab name, and a user profile for Dharshan A N. The main workspace is divided into three sections: a sidebar on the left with a list of questions (Q4, Q5) and their marks, a central area for writing and running code, and a right sidebar for input and output.

Library Management System
10 marks

Q4
Employee Data Processing
10 marks

Q5
Online Voting System
10 marks

5/5 answered

Display all books

Your Code

```
1 import java.util.*;
2 class Main {
3     public static void main(String[] args) {
4         HashMap<String, String> books = new HashMap<>();
5         books.put("101", "Java");
6         books.put("102", "Python");
7         books.put("103", "C++");
8         System.out.println("Book with ISBN 102:");
9         System.out.println(books.get("102"));
10        System.out.println("\nAll Books:");
11        Iterator<String> it = books.keySet().iterator();
12        while (it.hasNext()) {
13            String isbn = it.next();
14            System.out.println(isbn + " : " + books.get(isbn));
15        }
16    }
17 }
```

Input
stdin input

Output

```
Book with ISBN 102:
Python
All Books:
101 : Java
102 : Python
```

Run Save

94°F Mostly cloudy 13:42 06-08-2026

4. Suppose that a company processes employee records and filters employees based on salary.

(a) List three advantages of Generics in Collections.

(b) Represent how employee data is stored using List with Generics.

(c) Write a Java program using Iterator to:

. Store employee objects

. Display employees with salary > 50,000

The screenshot displays the SIMATS Online Programming Lab interface. The browser address bar shows the URL: `simatslab.saveetha.com/practice_app/classactivity/attempt/assignment_id=747`. The page header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", and the user profile "Dharshan A N" with ID "192525370RA".

On the left sidebar, there are two questions listed:

- Q4 Employee Data Processing (10 marks)
- Q5 Online Voting System (10 marks)

Below the questions, it says "5/5 answered".

The main area is titled "Your Code" and contains a Java program:

```
1 import java.util.*;
2 class Main {
3     public static void main(String[] args) {
4         ArrayList<String> emp = new ArrayList<>();
5         emp.add("101 Ram 60000");
6         emp.add("102 Ravi 45000");
7         emp.add("103 Priya 70000");
8         Iterator<String> it = emp.iterator();
9         System.out.println("Employees with Salary > 50000");
10        while (it.hasNext()) {
11            String s = it.next();
12            String[] data = s.split(" ");
13            int salary = Integer.parseInt(data[2]);
14            if (salary > 50000)
15                System.out.println(s);
16        }
17    }
18 }
```

Below the code editor are "Run" and "Save" buttons.

On the right side, there is an "Input" field labeled "stdin input" and an "Output" field showing the results:

```
Employees with Salary > 50000
101 Ram 60000
103 Priya 70000
```

The bottom of the screen shows a Windows taskbar with the date "06-08-2026" and time "13:42".

5. Suppose that an online voting system ensures one vote per user and counts votes efficiently.

(a) Identify three suitable Collection types for this system.

(b) Design a structure using Set and Map for voters and vote counting.

(c) Write a Java program using HashSet and HashMap to:

. Record votes

. Prevent duplicate voting . Display results

The screenshot displays the SIMATS Online Programming Lab interface. The browser address bar shows the URL: `simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=747`. The page header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", and the user profile "Dharshan A N" with ID "192525370KA".

On the left sidebar, there are two questions listed:

- Q4 Employee Data Processing (10 marks)
- Q5 Online Voting System (10 marks)

Below these, it indicates "5/5 answered".

The main area contains a Java code editor with the following code:

```
1 import java.util.*;
2 class Main {
3     public static void main(String[] args) {
4         HashSet<String> v = new HashSet<>();
5         HashMap<String, Integer> m = new HashMap<>();
6         m.put("A", 0);
7         m.put("B", 0);
8         String id = "U101";
9         String c = "A";
10        if (!v.contains(id)) {
11            v.add(id);
12            m.put(c, m.get(c) + 1);
13            System.out.println("Vote Recorded");
14        } else {
15            System.out.println("Duplicate Vote");
16        }
17        id = "U101";
18        c = "B";
19        if (!v.contains(id)) {
20            v.add(id);
21            m.put(c, m.get(c) + 1);
22        } else {
23            System.out.println("Duplicate Vote");
24        }
25        Iterator<String> i = m.keySet().iterator();
26        System.out.println("\nResults:");
27        while (i.hasNext()) {
```

On the right side, there is a "stdin input" field and an "Output" window. The output window shows the following text:

```
Vote Recorded
Duplicate Vote
Results:
A : 1
B : 0
```

The bottom of the screen shows a Windows taskbar with the date and time "13:42 06-08-2026" and system status icons.